Master prompt: build the exam engine

VOLUME 02 / BUILDER INSTRUCTION

Emergent, implement a dependable exam engine for an independent practice platform serving Tejas and a growing student population. Treat the following pages as executable product requirements. Use the approved 2026 configurations as baselines; require a separate evidence review before advertising any later year as an official simulator.

Build the shared core

Create one reusable attempt service with exam-specific policies for content structure, answer types, navigation, timing, marking, and result release. Keep the scoring function independent of React components. The server owns the accepted answer ledger, attempt deadline, paper lock, and final score. The browser renders an accessible blue interface, stages edits, and sends explicit commands. Display an independent-practice label wherever a student might mistake the platform for an examination authority.

Implement the smallest complete path first: publish a reviewed paper, start an attempt, save answers, recover a refresh, submit, calculate the score, and open the question-wise solutions. Make this path work with actual database persistence before adding advanced analytics. Reuse the approved content and solution contracts from Volume 03 and the measurement definitions from Volume 04.

Preserve exam distinctions

Support the JEE Main Paper 1 integer-entry contract, the BITSAT optional extra-question lock, the two WBJEE paper sessions with category scoring, and the COMEDK single-choice contract. WBJEE is an OMR examination; label its online experience a digital adaptation rather than an official CBT clone. Keep training conveniences, such as pause and hints, in a separate mode with a separate comparison cohort.

Deliver concrete evidence

Return the policy registry, schema migrations, attempt and answer command APIs, deterministic scoring fixtures, and runnable end-to-end checks. Provide an administrator preview that uses the exact released configuration. Add a visible readiness report listing verified source documents, unresolved rules, content review status, and failed checks. A paper with unresolved format or scoring requirements must remain unpublished as an official-style simulator. Label comparative scores Mock percentile within this paper's platform cohort. Never call them an NTA Score, official normalized percentile, or official admission rank; this platform does not observe the authority's full examination population.

ACCEPTANCE / PRECAUTIONS

- Demonstrate a refresh with no loss of committed answers and no added time.
- Show the BITSAT base-paper lock and a WBJEE fractional-score example.
- Keep COMEDK duration provisional until the timing evidence gate is passed.

Rule registry and year-specific release gates

POLICY GOVERNANCE

Make examination behavior data-driven. A school paper may be educationally useful without matching an official examination format. Expose that distinction before the student starts, and store the selected policy version so the resulting score remains reproducible months later.

Define the policy record

Create ExamPolicy fields for examCode, variantCode, examYear, policyVersion, status, sourceIds, sourceDocumentHash, reviewedAt, reviewedBy, timingEvidenceStatus, and policyHash. Separate machine-readable behavior from student-facing instructions. Support statuses draft, evidence_pending, verified, approved, retired, and superseded. A verified source is not automatically an approved software configuration; a reviewer must confirm that the configured values correctly represent that source.

Use a field-level evidence ledger. Record which source supports duration, section counts, marking, numeric input, navigation, review handling, accommodations, and disputed-question treatment. Mark a missing field unresolved instead of filling it from a familiar prior year. When a notice changes one field, preserve the old policy and create a new version with a precise diff.

Separate three product modes

Offer official-style simulation, digital adaptation, and custom training. Show a prominent mode badge and describe departures in plain language. A custom integer-only JEE practice paper must not silently inherit a full JEE Main claim. A compressed WBJEE session must not be presented as a full two-paper rehearsal. Allow administrators to configure educational variants while keeping their analytics cohorts distinct.

Prevent stale release claims

Do not infer 2027 rules from the retrieved 2026 bulletins. The release gate must require the relevant official document or approved instructions for the selected year. Display the verification date in the administrator report and a concise baseline year to students. For sources whose publication date is not explicit, identify the document year and retrieval date without inventing an issuance date. Treat active attempts as immutable consumers of their starting policy.

ACCEPTANCE / PRECAUTIONS

- Changing a policy must never change an active attempt.
- A missing timing source prevents the verified-simulator badge.
- Reject publication if subject totals, question counts, or maximum marks disagree.

JEE Main Paper 1 configuration

VERIFIED 2026 BASELINE

Implement the engineering Paper 1 variant only in the initial JEE release. The NTA 2026 bulletin linked from its official site is the source of this baseline; the source file is in its November 2025 upload path and was checked on 11 October 2026. [S2-01]

Configure the published structure

Set three subjects: Mathematics, Physics, and Chemistry. Each has 20 single-choice questions and five numerical-answer questions. The paper has 75 questions, maximum 300 marks, and a standard duration of three hours. For both question types, correct earns four, incorrect loses one, and unanswered earns zero. The numerical answer is entered as an integer rounded to the nearest integer. [S2-01]

Express the structure through sections and item groups; do not calculate marking from the position of a question in an uploaded PDF. Each item must carry an approved type and scoring policy. If a PDF contains additional optional numerical questions from a different year, block the 2026 template assignment until the administrator selects a matching historical or custom policy.

Build the preflight validator

Validate subject counts, item-group counts, available options, integer keys, supported language assets, complete solution mapping, and total maximum score. Show each mismatch with the affected question identifier and remediation. An administrator must choose either to fix the paper or publish it as custom practice. Do not quietly truncate, duplicate, or manufacture questions to meet a target count.

Keep additional variants explicit

Do not expose B.Arch or B.Planning tabs inside this Paper 1 implementation. Their content and assessment requirements require separate policies and acceptance criteria. If multilingual content is offered, retain an explicit source-language and translation-review status for every item. Provide the candidate's approved language assets consistently throughout the attempt, including diagrams and instructions. Log the language choice with the attempt snapshot.

ACCEPTANCE / PRECAUTIONS

- A complete standard paper validates against the 2026 subject structure.
- An integer-only teacher paper appears as custom practice.
- A paper with a missing solution can be rehearsed only if the release policy explicitly allows delayed solutions.

JEE numerical input and key precision

INTEGER ANSWERS

Implement integer entry as a distinct answer component. A numerical response must not behave like a multiple-choice item or a calculator. Keep the student's raw input, validated canonical value, and final committed value separate for debugging and question-wise review.

Build a deliberate input contract

Use an on-screen numeric keypad and a keyboard-accessible equivalent. Permit the characters supported by the approved policy, including a leading minus sign when negative integer answers are valid. Reject decimals, exponent notation, unit strings, multiple signs, infinity, and empty sign-only values in the standard integer-entry variant. Make zero a valid value distinct from an unanswered field. Show an inline error near the input without erasing the student's draft.

Normalize harmless formatting such as leading zeros only after preserving the original draft for diagnostic purposes. Store the accepted integer as an exact decimal string or a suitably bounded integer; never run arbitrary expressions supplied by the user. Avoid silently rounding a student's decimal input into a scoreable response. The candidate performs the calculation; the platform records the submitted integer.

Review mathematical keys

Keep an exact reviewed integer key and a separate authoring note containing the underlying computed value where useful. The source requires nearest-integer rounding but does not fully define a software tie-breaking function for every half-integer edge. Flag authoring keys that depend on an exact half-integer convention for a subject expert. Do not use JavaScript rounding as an unreviewed mathematical authority. [S2-01]

Test response semantics

Create fixtures for blank, zero, positive, negative, leading-zero, out-of-range, decimal, and invalid-character drafts. Ensure Clear Response empties the editing field, and explicit save commits the resulting blank under the selected navigation contract. In review, show the final committed response and the key, with discarded drafts confined to an optional technical history. Keep input length and domain constraints in the versioned policy so future variants can support decimal answers deliberately.

ACCEPTANCE / PRECAUTIONS

- Blank and zero produce different answer states.
- A decimal draft cannot silently become an integer answer.
- A reviewer must resolve a half-integer ambiguity before publication.

JEE palette and review behavior

SAVE SEMANTICS

Reproduce the decision consequences of the published CBT instructions. The NTA appendix says a palette jump does not save the current answer; Save & Next and Mark for Review & Next save it. An answered-and-reviewed response is evaluated, while an empty reviewed item is not. [S2-01]

Separate draft from committed response

Store draftAnswer in the editing and optional durable recovery context, never in the grading ledger. A recovered draft is not a committed answer. Store committedAnswer in the accepted ledger. Add visited and markedForReview as independent fields. An option click changes the draft. Save & Next commits that draft, clears the review flag, and navigates. Mark for Review & Next commits the current draft, sets review, and navigates. A palette or section jump follows the policy's discard-draft navigation path. Preserve the previously committed answer when an unsaved replacement is discarded.

Clear Response changes the editable draft to blank. The next explicit save commits blank. Verify the final Clear Response behavior against the approved interface evidence before claiming exact visual parity; include a visible unsaved-change indicator in the platform adaptation. Treat Save and Mark actions as acknowledged commands, not mere changes of button color.

Render five understandable states

Derive not visited, visited without answer, answered, reviewed without answer, and answered plus reviewed from accepted state. Pair each palette color with a shape or icon and readable legend. The current question highlight is a separate visual layer. A pending save badge must not masquerade as a confirmed answered state. Show consistent section counts derived from the same accepted ledger used by submission.

Avoid accidental score changes

Do not score the review flag itself or add a penalty for revisiting. Keep a candidate's navigation available within the permitted phase. In the submission summary, state which saved answers are counted and how an empty reviewed question is treated. An optional training mode may autosave on navigation, but it must declare that behavior and remain outside the official-style comparison cohort.

ACCEPTANCE / PRECAUTIONS

- Changing an answer and jumping away preserves the earlier committed answer.
- An answered-and-reviewed correct response earns the same marks as a saved correct response.
- The submission summary and palette use the same authoritative counts.

BITSAT base-paper experience

VERIFIED 2026 BASELINE

Use the retrieved official BITSAT-2026 brochure for the engineering Mathematics variant. The brochure's document year is explicit; an original publication date is not asserted here. Record the retrieved document hash and verification date of 11 October 2026. [S2-02]

Configure the base session

Set 180 minutes without a break and 130 four-option, single-correct questions: Physics 30, Chemistry 30, English Proficiency 10, Logical Reasoning 20, and Mathematics 40. Correct earns three, incorrect loses one, and unanswered earns zero. There is no separate time limit for individual parts, and candidates may revisit the base questions before accepting the extra-question phase. Instructions and questions are in English. [S2-02]

Create the subject navigation in this recognizable order, with English and Logical Reasoning distinct in both reports and counters. Do not merge them into a generic aptitude score. Show base-paper completion independently of correctness, since eligibility for extras depends on saved responses rather than the answer key. Keep correctness hidden during the running attempt. Derive maximums from this policy: 390 for the base, 36 for extras, and 426 combined when the extra phase is entered. For product score percentages, use 390 without entry and 426 after entry; show base and extra contributions separately. These are platform scoring percentages, not official percentiles.

Prepare a complete paper

Validate the uploaded question set against the approved subject allocation. Require all optional extra-question assets to be present and reviewed before enabling that facility. A paper that lacks extras may still be offered as base-only custom practice, but the start page must say so. Do not expose an extra-question button that leads to generated placeholders or randomly assembled unreviewed content.

Keep mode variants isolated

Treat a Biology path as a separately approved content variant with its own subject allocation and labels; do not switch Mathematics to Biology after the attempt begins. Keep the core engine capable of that future variant without expanding the launch scope by accident. The candidate's selected variant, extra-question availability, navigation policy, and solution release policy belong in the immutable attempt snapshot. Results must distinguish base score, extra score, and combined raw score.

ACCEPTANCE / PRECAUTIONS

- Base completion counts accepted answers, including answered-and-reviewed items.
- English and Logical Reasoning have separate analysis identifiers.
- A missing extra-question set disables the feature before the student starts.

BITSAT extra-question transaction

IRREVERSIBLE PHASE LOCK

Implement the extra phase as a guarded server transaction. Under the 2026 brochure, it is optional only after all base questions are answered and time remains; accepting it prevents returning to correct the original answers. The extra set has three questions each from Physics, Chemistry, Mathematics, and Logical Reasoning. [S2-02]

Show an honest decision dialog

Offer a restrained button only when the server confirms eligibility. The dialog must say that the base answers will be locked, the existing timer continues, and the extra phase retains negative marking. Show accepted base completion and remaining server-derived time. Do not recommend entering merely because the button is available. This choice is a strategy decision with a risk of reducing the combined score.

Require a clear confirmation label such as Lock base answers and enter extras. Place Cancel with equivalent readability. Disable repeated clicks while the request is pending. If the dialog is open at expiry, show the final submission state; it must not grant a new phase or extra time.

Make the transition atomic

Process ENTER_EXTRA with an idempotency key, expected attempt revision, and latest accepted command sequence. Lock the attempt, check every base item has a committed response, verify the phase and deadline, freeze the base ledger hash, and transition once. Fetch extra content only after authorization succeeds. Do not trust a browser-supplied completion count or unlock flag. A retry returns the same accepted transition rather than making a second extra phase.

Preserve the locked contract

Reject all later base-answer edits server-side, including old queued commands from another tab. Render locked base items read-only or remove their navigation during the phase. Keep the frozen base answers available in the final review. Extras do not reset the attempt deadline. A refreshed browser must recover the locked phase and remaining time. Track entry time and base snapshot hash for diagnostics and later strategy analysis without inventing official extra-phase percentiles. Preserve every untouched locked base contribution in the combined result, including its negative marks; entering extras does not replace the base score.

ACCEPTANCE / PRECAUTIONS

- A missing base response makes entry fail even if the browser claims full completion.
- A queued base edit arriving after the lock is rejected.
- Entering extras, refreshing, and retrying entry never unlocks the original paper.

WBJEE two-paper scheduling

VERIFIED 2026 BASELINE

Model WBJEE as two related paper attempts rather than one interchangeable subject session. The official 2026 bulletin in the 10 March 2026 upload path describes an OMR examination and separate scheduled papers; it was verified on 11 October 2026. [S2-03]

Represent the two papers accurately

Paper I is Mathematics: 75 questions, maximum 100 marks, with a two-hour session. Paper II combines Physics and Chemistry: 40 questions and 50 maximum marks per subject, also two hours. The reviewed 2026 reference schedule is 11:00-13:00 for Mathematics and 14:00-16:00 for Physics-Chemistry, a one-hour interval in Asia/Kolkata. Keep the rehearsal date configurable. Together, the papers contain 155 questions and 200 maximum marks. An engineering rehearsal therefore needs both paper scores; completing one paper must not silently stand for the full examination. [S2-03]

Create an ExamSeries record that links two independent attempts through a shared student and paper-package version. Store separate startedAt, deadlineAt, submission state, accommodations, responses, and scoring versions. One paper expiring must not close or reopen the other. A student can review scheduling instructions before starting either session without consuming answering time.

Offer honest rehearsal choices

Provide full-day rehearsal, two-paper flexible rehearsal, and individual-paper training. In full-day rehearsal, enforce the configured sequence and interval. In flexible rehearsal, allow the student to schedule the papers independently but clearly label that departure. Record mode in analytics so split rehearsal results are not mistaken for same-day performance. A shortened interval is a custom setting, not an official recommendation.

Release results with correct scope

For the product's engineering practice report, aggregate the two linked raw paper scores only after both are submitted. Show pending second paper instead of a fabricated full total. A paper-only report is still useful for topic feedback but should not display a full engineering mock rank. Keep a future pharmacy-report variant separate because the relevant official ranking basis differs from the engineering basis. Do not implement admission eligibility decisions in the exam engine.

ACCEPTANCE / PRECAUTIONS

- The Mathematics timer is independent from the Physics-Chemistry timer.
- An incomplete series displays paper-only performance.
- Full-day and flexible-rehearsal results belong to separate comparable cohorts.

WBJEE categories and rational scoring

PARTIAL CREDIT ENGINE

Implement category-aware scoring from the official 2026 bulletin. Mathematics category counts are 50, 15, and 10; Physics and Chemistry each use 30, five, and five. Keep category on every item rather than deriving it from the display order. [S2-03]

Encode the scoring policies

Category 1 awards one for correct, minus 0.25 for incorrect; Category 2 awards two for correct, minus 0.5 for incorrect. Both are single-correct, with a multiple-option response treated as incorrect. Category 3 awards two for all correct options only; a proper correct subset receives two times selected-correct count divided by total-correct count. Any incorrect option makes that response worth zero; blank is zero. [S2-03]

Represent Category 3 keys and answers as option-ID sets. Score by set inclusion and cardinality, not string equality of concatenated option letters. The same logical response must score identically after any approved display permutation. Preserve the published option identities in the review view so a source solution remains intelligible.

Preserve exact fractional totals

Use a rational numerator and denominator, or a documented high-precision decimal contract, for intermediate partial credits. Do not round every item to two decimals before summing. A two-thirds item and a four-thirds item must aggregate to two exactly. Define final display precision separately from the comparison value. Include a short marks explanation in each question-wise review, including the chosen set, correct set, and partial-credit calculation.

Validate potentially invalid responses

For the ordinary digital Category 1 and 2 view, use radio buttons and prevent multiple selection. For imported OMR responses or an explicitly configured bubble simulator, retain malformed or multiple-bubble responses and apply the appropriate category outcome. Category 3 uses checkboxes and a label saying one or more answers may be correct. Never apply a generic all-or-nothing checkbox score to a proper correct subset.

ACCEPTANCE / PRECAUTIONS

- Key A,C with answer A earns one; answer A,B earns zero in Category 3.
- The answer and key sets remain stable if visible option order changes.
- Exact aggregate comparisons must not depend on two-decimal display rounding.

WBJEE digital adaptation and OMR rehearsal

MODE HONESTY

The official WBJEE bulletin describes OMR response marking that cannot be changed or erased. An editable browser interface is therefore an adaptation. Offer a useful digital experience while making its relationship to the real examination clear before any scored attempt begins. [S2-03]

Design the standard digital adaptation

Use an accessible question palette, paper timer, category badge, readable diagrams, and appropriate single-choice or multi-select controls. Permit response revision until submission in the standard learning adaptation. Show Digital WBJEE practice prominently and explain that the interface supports editable responses unlike the physical OMR sheet. Do not advertise this interface as an official WBJEE CBT replica.

Keep category warnings near the options: the multi-select category has a different scoring rule, and including a wrong option destroys its partial credit. Provide a paper-wide answer summary without indicating correctness. For paper exercises, allow a downloadable blank response-sheet layout only if its format is approved and clearly branded as a practice sheet.

Offer a separate irreversible-bubble mode

Implement OMR rehearsal as an optional separate mode. Stage a choice, ask for a final fill action, then freeze the selected bubble set for that question. For a multi-select item, permit the student to build the full draft set before committing it. An irreversible commit needs clear explanation and keyboard-accessible confirmation. Do not erase an accepted bubble because the student changes a local checkbox.

Compare like experiences

Persist the editing policy and OMR rehearsal flag with the attempt. Show the policy in the result context and separate its analytics cohort from editable digital practice. Log changes and final fills as different events; an editable practice change is a learning signal, not evidence that the student can revise a real OMR response. Provide a concise rehearsal note about checking question numbering and avoiding rushed transfers, derived from observed errors rather than alarming generic warnings.

ACCEPTANCE / PRECAUTIONS

- Editable practice and irreversible-bubble rehearsal have different mode badges.
- A committed bubble set cannot be changed by refresh or a second tab.
- Category 3 permits selecting the full intended set before the irreversible commit.

COMEDK configuration and evidence gap

PARTLY VERIFIED 2026 BASELINE

Use COMEDK's official Information Brochure 2026, notified on 3 February 2026, and its current examination page. The retrieved primary sources confirm structure and marking but do not explicitly state the 2026 duration. Treat that missing timing evidence as a real release gate. [S2-04] [S2-05]

Configure verified behavior

The published engineering pattern is 60 Physics, 60 Chemistry, and 60 Mathematics single-choice questions: 180 questions and 180 maximum marks. Each correct answer earns one, with no negative marking; the medium is English. The official CBT guidance includes a timer, palette, review, question-paper view, and movement between sections. [S2-04]

Use the common single-choice component but select a COMEDK scoring policy explicitly. Wrong and unanswered responses have different attempt states even when both receive zero. That distinction matters for accuracy, unattempted opportunities, and question-wise analysis. Do not reuse JEE negative marking merely because the blue CBT shell looks similar.

Handle duration responsibly

Provide 180 minutes as a provisional custom-mock default only, with timingEvidenceStatus unresolved. Before publishing a verified 2026-style simulator, obtain and review official timing instructions or an appropriately redacted official Test Admission Ticket. The release report must identify the reviewer and supporting timing source. Do not use an old-year important-dates page as current evidence, and do not silently upgrade a provisional default to verified.

Control unsupported facilities

Keep any virtual calculator disabled unless current primary official instructions explicitly authorize one. Third-party mock claims are insufficient. Let students use approved rough-work conventions without supplying a calculation aid in the running interface. Keep review and save semantics documented in the platform instructions and verified against available official interface evidence before claiming exact parity. The real exam's rank or normalized percentile is not reproduced from this platform's independent practice cohort.

ACCEPTANCE / PRECAUTIONS

- A COMEDK wrong response receives zero without becoming unanswered.
- A verified-simulator release is blocked while duration remains unresolved.
- No virtual calculator appears without an approved current primary source.

Configuration schema and validators

MACHINE-READABLE CONTRACT

Implement typed JSON policies with schema validation on both authoring and release. Configuration should describe the examination clearly enough that another developer can inspect it without searching application components for hidden duration, marking, or navigation constants.

Define core fields

Use examCode, variantCode, examYear, policyVersion, sourceIds, verificationStatus, deliveryMode, durationSeconds, timingEvidenceStatus, languages, sections, answerPolicies, scoringPolicies, navigationPolicy, reviewPolicy, submitPolicy, accommodationsPolicy, extraPhasePolicy, and releasePolicy. In each section, define subjectId, questionGroups, requiredCount, maximumMarks, orderPolicy, and scopeId. A group specifies answerType, category, scoringPolicyId, and count.

For a scoring policy, store correctAward, incorrectPenalty, blankAward, partialRule, acceptedKeyRepresentation, invalidResponseRule, droppedItemRule, allOptionsCorrectRule, and roundingDisplayPolicy. Give every exceptional rule an explicit unset or requires_review value if the primary source does not establish it. Never interpret an absent field as a permissive default.

Specify behavior rather than UI guesses

Let navigationPolicy define whether palette navigation saves, whether section jumps save, whether drafts are discarded, and which phases permit revisiting. Let extraPhasePolicy define eligibility scope, remaining-time requirement, content allocation, base lock, and shared deadline. Let submitPolicy define early-submit availability, confirmation, draft handling, cut-off relation, retry idempotency, and release timing. Blue styling belongs in design tokens rather than policy logic.

Enforce cross-field consistency

Reject negative durations, noninteger group counts, duplicate section IDs, unknown policy references, unsupported answer types, illegal partial rules for single-choice groups, and inconsistent maximum marks. Verify an extra phase references a real reviewed item set. Ensure every required policy field has evidence or is openly marked a custom training decision. Store the canonical JSON hash at approval. Provide an administrator-readable validation report with field paths and actual versus expected values, then rerun it whenever the paper manifest changes.

ACCEPTANCE / PRECAUTIONS

- A JSON policy can be validated without rendering the exam UI.
- Unknown or unresolved rules block the appropriate release claim.
- The published paper manifest and policy hash must be stored together.

Immutable attempt and content snapshots

REPRODUCIBLE RESULTS

Freeze the paper, policy, and presentation contract when an attempt starts. Every review screen must reconstruct what that student actually saw, even after an administrator corrects a solution, revises a tag, or publishes a new question version.

Persist the starting snapshot

Store attemptId, studentId, seriesId, examPolicyVersion, policyHash, paperVersion, paperManifestHash, itemVersionIds, sectionMapping, keyVersion, solutionVersion, languageVariant, displayPermutation, mode, accommodationsVersion, startedAt, deadlineAt, and releasePolicyVersion. Store stable internal option IDs separately from their visible letters. A student's visible choice B must always map to the same internal option used during scoring, even when another attempt displays a different order.

Pin content assets by versioned object key or immutable checksum. Updating an image URL in an administrator editor must not replace the diagram in an active attempt. Include the approved original question crop or source reference in the review manifest. Make all published paper manifests readable to authorized maintainers and keep answer keys restricted to scoring and reviewed solution release paths.

Snapshot answers at closure

At submission, freeze the accepted response ledger, its hash, final sequence watermark, phase-lock state, and completion reason. Generate a ScoreRun record containing scorerVersion, policyVersion, keyVersion, responseSnapshotHash, scoreBreakdown, and createdAt. Treat the score as a versioned output. Do not overwrite it in place after correcting a key; create a new score run linked to the reason for rescoring.

Handle corrections transparently

If an approved correction affects completed attempts, calculate the affected population before applying the new run. Display the previous and revised raw scores with a concise explanation in the student's report. Rebuild comparable analytics cohorts consistently, and mark their calculation date. Keep active attempts on their starting paper contract unless a documented incident policy deliberately closes or replaces them. Do not make a teacher's late edit silently alter the student's answer history.

ACCEPTANCE / PRECAUTIONS

- A historical review reconstructs the exact item and option order shown.
- Changing a solution creates a new version rather than mutating a completed score.
- Scoring reads a frozen accepted ledger, never the latest editable paper.

Attempt lifecycle as a state machine

ALLOWED TRANSITIONS

Implement one explicit lifecycle reducer on the server. Route guards and UI screens must consume its state instead of inferring progress from localStorage, the existence of a timer, or a client-side score value.

Declare the states

Use CREATED, PREFLIGHT, READY, RUNNING_BASE, RUNNING_EXTRA, SUBMITTING, SUBMITTED, SCORING, SCORED, RELEASED, and VOIDED. Allow CREATED to PREFLIGHT after authorization; PREFLIGHT to READY after requirements pass; READY to RUNNING_BASE only through START. START sets the authoritative start and deadline exactly once. RUNNING_BASE to RUNNING_EXTRA is available only under the approved extra-phase policy. Either running state may close through user submission or deadline expiry.

SUBMITTING freezes the response cutoff and gathers accepted records through the required watermark. SUBMITTED contains the final immutable response snapshot. SCORING produces a versioned score; SCORED awaits the configured result condition; RELEASED means the result is eligible for display. Keep result_release_status and solution_release_status as independent server gates. A raw score may appear while explanations remain pending under a solution embargo. RELEASED never automatically grants solution access. VOIDED records a documented operational or integrity decision with a reason, reviewer, timestamp, and audit reference; it must not mean deleted.

Keep connectivity orthogonal

Do not make DISCONNECTED a lifecycle state that resets timing. Connectivity is an independent client status: online, degraded, offline, reconnecting. A running attempt remains running during a network loss. Likewise, focus, fullscreen status, current question, and writer-device ownership are separate attributes. The state machine should remain understandable when multiple symptoms occur at once.

Constrain transitions precisely

Require authorization, an idempotency key, expected revision, and policy guards for every transition. A second START returns the existing attempt contract; it cannot create a later deadline. A repeated SUBMIT returns the same final receipt. A request to reopen a submitted attempt fails unless an authorized incident process creates a separate replacement or documented adjustment. Recover a stuck SCORING operation with an idempotent retry; do not ask the student to retake the paper because an analytics worker crashed.

ACCEPTANCE / PRECAUTIONS

- Test every allowed transition and each forbidden adjacent transition.
- Refresh never creates a new START or changes the deadline.
- A scoring retry cannot duplicate the attempt or its final receipt.

Question-state reducer and command semantics

AUTHORITATIVE ANSWERS

Implement a deterministic question reducer that derives status from data. Keep answer content, visit history, review intention, local edit state, and network acknowledgement as distinct concepts so the interface can explain every state accurately.

Define the stored fields

For each item, keep visited, committedAnswer, reviewFlag, questionRevision, lastAcceptedCommandId, committedAt, and phaseLock. Keep draftAnswer, dirty, saveStatus, and draftUpdatedAt in the editing context. Derive NOT_VISITED when never visited; VISITED_EMPTY when visited with blank committed answer and no review; ANSWERED when committed answer exists and no review; REVIEW_EMPTY when blank and reviewed; ANSWERED_REVIEW when answered and reviewed. Zero and a nonempty option set count as answers.

VISIT marks visited without changing a response. SELECT replaces the draft for radio items or updates the draft set for checkboxes. TYPE edits a numeric draft. CLEAR empties the draft. SAVE commits the validated draft and removes review under the selected policy. MARK_REVIEW commits the draft and sets review. JUMP discards an unsaved draft when the selected exam policy requires that behavior. Explicitly define each event in code and documentation.

Validate every answer command

Check that the attempt is running, the writer lease is valid, the question belongs to this snapshot, the phase permits editing, and the answer conforms to the type. Reject unknown option IDs, duplicate IDs, out-of-domain numeric strings, and edits to locked base questions. Validation failure must preserve the draft with a useful correction message. A stale revision returns the latest accepted state and a conflict code rather than overwriting a newer answer.

Project state consistently

Derive palette icons, section counters, submission summaries, and review labels from the same accepted fields. Overlay a pending-save marker without falsifying committed status. Emit compact answer-history events for later change analysis, but score only the final accepted response. Serialize state predictably so replaying the accepted commands reproduces the snapshot hash. Avoid a separate UI-only answered boolean that can drift from the stored answer.

ACCEPTANCE / PRECAUTIONS

- Replaying accepted commands reproduces the same final question state.
- A pending save is visibly distinct from a committed answer.
- A blank review state cannot accidentally become an attempted question.

Server clock, deadline, and timer display

TIME IS A CONTRACT

Use one authoritative time contract per paper attempt. A browser countdown is a display of that contract, not the mechanism that grants or removes time. The student must never gain minutes by changing the device clock, refreshing, sleeping the phone, or opening another tab.

Establish time once

At START, store serverStartedAt, approvedDurationSeconds, accommodationsAdjustmentSeconds, and deadlineAt. For scheduled tests, explicitly define whether a hard close truncates late starts or whether each admitted student receives the full duration. Show that rule before starting. Instruction-reading time should be outside the answering window unless the declared custom policy deliberately includes it. Starting an attempt must be a server transaction that returns the existing record on retry.

Return serverNow and deadlineAt on start, resume, and heartbeat. Calculate an initial display offset, then use a monotonic browser timer for smooth countdown between synchronizations. Periodically reconcile with server time without letting the display jump backward into extra time. Clamp a visually expired timer at zero, while the server independently performs closure and rejects late edits.

Do not pause on connectivity loss

For strict exam practice, wall-clock time continues during offline periods. State this on the start page and offline banner. A pauseable training mode needs a separate elapsed-time model and a different analytics cohort. Operator incident extensions, if enabled, require an authorized recorded command with reason and new deadline; a client retry or support refresh is insufficient.

Define expiry and boundary behavior

Use a documented exclusive cutoff: a new answer command is eligible only when its trusted server receipt time is before deadlineAt. Commands at or after the deadline are rejected as late. Align the worker's closure query with the same inequality. Warn at configured thresholds through calm text and an accessible optional sound. A warning is a product convenience, not permission to change the examination duration.

ACCEPTANCE / PRECAUTIONS

- Advance the client wall clock by a day; accepted remaining time stays unchanged.
- Suspend the browser past expiry; resume opens the final state.
- Test commands just before, exactly at, and just after the cutoff.

Save queue and reconnection protocol

LOSS PREVENTION

Build a small durable client journal for the student's own responses and commands. It is a recovery aid, not an authority that can override server deadlines or a substitute for acknowledging that an answer was saved.

Specify the acknowledgement contract

Assign each answer command a UUID idempotency key, deviceId, writer fencing token, questionId, expectedQuestionRevision, local sequence, and payload hash. Queue commands locally in order and send them promptly. Return acceptedCommandId, serverReceiptAt, newQuestionRevision, acceptedAnswer, reviewFlag, and attemptPhase. Display Saved only after this acknowledgement. Show Saving, Offline: pending, or Save rejected when those states actually apply.

Use one authoritative writer lease per attempt. A second tab may observe state but must explicitly take over the lease before editing; takeover invalidates the old fencing token. After takeover, the old device's queued commands fail safely. Local sequence assists ordering within a device but does not authorize a stale device or turn a client timestamp into trusted evidence.

Resume from the server snapshot

On reconnect, request lifecycle, phase, deadline, question revisions, and accepted command identifiers. Deduplicate journal entries already acknowledged by the server. Replay remaining commands only when the attempt is still running, their lease is valid, their target is unlocked, and their revisions are compatible. Show an explicit conflict resolution screen when a later accepted edit exists; do not silently prefer whichever request arrived last.

Handle practical network failures

Use bounded exponential backoff with jitter and retry only safe idempotent operations. Keep pending response data across a refresh where storage is available, while clearing it after final reconciliation under the retention policy. If storage is unavailable, warn before the attempt and keep server saves frequent. Offline commands arriving after the strict cutoff are not scored, even if the client claims they were made earlier. Explain that consequence calmly and preserve a support trace without falsifying the result.

ACCEPTANCE / PRECAUTIONS

- Lose the acknowledgement, retry, and receive the same accepted command result.
- A second device cannot overwrite answers using an expired writer token.
- Reconnecting after expiry shows rejected pending edits and the final receipt.

Submission and deadline race resolution

EXACTLY ONE FINAL SNAPSHOT

Make submission a convergent server process. A student clicking Submit at the same moment the deadline worker runs must produce one response snapshot, one completion reason, and one receipt, regardless of request order or transport retries.

Define the ingress boundary

Record command receipt durably at a trusted server boundary and bind that timestamp to the authenticated attempt. Do not use a client-supplied editedAt value for deadline eligibility. If workers process requests asynchronously, eligibility may depend on this trusted receipt timestamp, but finalization must wait for the pre-cutoff ingress watermark to drain. A command merely sitting in the browser is not received.

For early submission, freeze a cutoff at the authorized submit transaction and ensure eligible commands preceding that cutoff are resolved before the final response hash is calculated. The UI should flush pending saves before asking for final confirmation. A user may choose to submit despite a displayed save failure, but the dialog must state that only accepted responses will count.

Resolve competing closures

Acquire an attempt-level lock or equivalent compare-and-swap. If already closing or closed, return its existing receipt. If time has expired, use deadline completion semantics even if the incoming request says user submission. If still running and the early-submit policy allows closure, freeze the user cutoff. Reject later answer mutations. Persist closure state and an outbox event atomically so a worker crash cannot strand an unannounced successful submission.

Return an auditable receipt

Include attemptId, completionReason, cutoffAt, acceptedAnswerCount, rejectedPendingCount where known, responseSnapshotHash, and resultStatus. Keep scoring separate from receipt issuance; show Result processing if the score is not ready. Poll or subscribe using the existing attempt, never create a new submission. The final score worker must use the frozen snapshot and be idempotent by snapshot hash plus scorer and key versions.

ACCEPTANCE / PRECAUTIONS

- Run submit and expiry concurrently in both orders; final responses match the declared cutoff.
- A pre-cutoff acknowledged command cannot disappear from the score snapshot.
- Duplicate submit requests return one receipt and one score run.

Navigation, display, and working-memory safety

EXAM SCREEN PARITY

Build an exam screen that keeps the question readable, the answer action clear, and the student's location obvious. Use a restrained blue shell, with the exam's actual behavioral policy determining navigation rather than a generic quiz layout.

Organize the desktop view

Place subject or section tabs above the question area, remaining time and attempt identity in a stable header, and the question palette in a collapsible side panel. Keep Save & Next, Clear Response, and Mark for Review actions within reach without obscuring long options or diagrams. Preserve the candidate's position in the question area when opening a lightweight help legend, then restore focus when it closes.

Show question number, subject, item category or answer type when relevant, and mark scheme in concise terms. Do not show chapter tags, difficulty hints, solutions, correctness, or cohort success rates while a strict attempt is running. Those details can influence answering strategy and belong in practice-mode scaffolding or after submission.

Protect question fidelity

Render reviewed text, mathematics, chemical notation, tables, and diagrams from approved assets. Preserve option labels and source numbering even if internal IDs differ. A shared passage needs stable access beside each linked item. A diagram zoom must not crop labels or reset the chosen answer. If an asset fails, retry from its immutable version and offer a report-issue control; do not invent a replacement question inside the running attempt.

Plan responsive and accessible behavior

On smaller screens, convert the palette to a drawer and keep a compact timer visible. Use full-width options and a touch-friendly keypad. Make the current question and save state available to screen readers. Use semantic radio and checkbox groups, visible focus, and text equivalents for icons. A full-question-paper view must remain read-only with respect to responses unless the policy explicitly permits embedded editing. Navigation should never unexpectedly scroll the student to a different item's answer field.

ACCEPTANCE / PRECAUTIONS

- A long stem and four long options remain readable at 200 percent zoom.
- Opening the palette or image zoom preserves the draft and focused control.
- Strict mode reveals no tags, keys, hints, or live success percentages.

Accommodations and accessibility variants

FAIR PRACTICE CONTRACTS

Implement accessibility as a supported capability of the core exam screen. Timing accommodations are a separate approved policy choice. Never infer extra time or an official entitlement from a self-selected visual preference or an unverified medical description.

Separate access features from time changes

Provide larger text, keyboard navigation, high contrast, reduced motion, screen-reader labels, accessible mathematical presentation where supported, and sufficient target sizes in every mode. These presentation settings do not change score or duration. Where a diagram needs an alternative description, ensure it conveys the reviewed information without adding a solution hint. Flag an image-only question with no meaningful alternative as an accessibility content defect.

Keep time accommodations as an explicit configuration with approval basis, scope, duration adjustment, policy version, and effective time. The NTA 2026 Paper 1 table includes four hours for the specified candidates eligible for a scribe; this is not a blanket extra-time rule for every person with a disability. [S2-01] Do not transfer that rule to BITSAT, WBJEE, or COMEDK without their own reviewed evidence.

Offer a practical learning option

Allow an extended-time training setting for students who want to practice comfortably, but label it custom practice and separate its percentile cohort. A student should not have to submit sensitive disability documents to select a private learning convenience. For a formally administered class test, store only the minimum approved accommodation record needed by the institution's workflow, following the platform's access and retention controls.

Validate the actual journey

Run start, answer, review, clear, navigate, reconnect, and submit with keyboard only and with a screen reader. Ensure the timer is not announced every second; provide milestone announcements and a manual remaining-time control. Do not require fullscreen or webcam permission in a way that blocks assistive tools. Persist the accommodation version in both linked WBJEE paper attempts and in every comparative report so timing context remains visible.

ACCEPTANCE / PRECAUTIONS

- Font size changes do not extend the timer.
- Custom extra time appears in result context and cohort filters.
- A keyboard-only student can complete the entire attempt without a pointer.

Scoring fixtures, anomalies, and rescores

DETERMINISTIC MARKS

Implement scoring as a pure, versioned function of approved policy, frozen question keys, and accepted responses. Return item-level awards and reasons so every mark in the total can be explained to the student and reproduced by a reviewer.

Create readable fixture cases

For ordinary items, test correct, incorrect, blank, reviewed-correct, and reviewed-blank responses. Under the named JEE 2026 policy, two correct and one wrong yield seven; under BITSAT's named base policy the same pattern yields five; under COMEDK's published marking it yields two. These are derived test expectations, not promised student performance. Cite the source policy in the fixture metadata. [S2-01] [S2-02] [S2-04]

For WBJEE Category 3, use key A,C: answer A,C earns two, A earns one, A,B earns zero, and blank earns zero. Add a three-correct-option case with one and two correct choices, verifying their exact fractions aggregate before display rounding. Test a Category 1 multiple-bubble imported response as incorrect even when it includes the correct bubble. [S2-03]

Represent exceptional key statuses

Use key statuses approved_single, approved_multi_valid_alternatives, all_options_correct, dropped, provisional, and disputed where appropriate. Encode the JEE 2026 branches explicitly: an MCQ with multiple valid alternatives awards four when the chosen option is among those alternatives; all options correct awards four only to attempted responses, with blank receiving zero; a dropped MCQ awards four to every participant in the relevant shift, including blank. A numerical item found invalid or dropped likewise awards four to every shift participant, including blank. [S2-01] Add fixtures proving the blank versus attempted distinction. A generic exclude-item-from-denominator remedy must not override these profile-specific awards. Do not transfer JEE remedies to another exam without reviewed evidence.

Control corrections and explanation

Block ordinary scoring on a provisional or disputed key unless an approved release policy explicitly permits a provisional report. Resolve corrections through subject review, publish a new key version, identify affected attempts, and create a rescore run. Keep the student's original accepted answers intact. Each award needs a reason code such as correct, incorrect, blank, partial_correct_subset, contaminated_subset, or dropped_credit, with its numeric contribution shown in the review. Check item sums against subject and total aggregates.

ACCEPTANCE / PRECAUTIONS

- Review status alone never changes the marks awarded.
- Every total equals the sum of its item-level contributions.
- A rescore creates a new result version and an explanation of the change.

Integrity signals and the limits of browser controls

INDEPENDENT PRACTICE

Protect the answer key and preserve honest comparison without pretending a home browser can reproduce a supervised test center. Browser signals can support review, but they are incomplete observations and cannot establish cheating by themselves.

Keep the server boundary firm

Authorize every content and command request against the student's attempt and release state. Do not send keys, solutions, hidden explanations, difficulty tags, or scoring scripts containing answers in the running payload. Cache only approved question assets for the current attempt. Protect teacher and administrator APIs separately from student APIs, and test object-ID substitution attempts across users and papers.

A signed asset URL can reduce casual sharing but cannot prevent screenshots or a student photographing the screen. Make that limitation explicit in the administrator product description. Avoid claims such as cheat-proof, guaranteed authentic score, or official test security. An independent practice result should primarily support learning.

Use proportionate integrity observations

If scheduled comparison tests require it, collect minimal events for visibility changes, repeated writer takeovers, implausible answer-rate bursts, and unusual response similarity. Record what was actually observed and its uncertainty. A tab switch may result from an assistive tool, network portal, incoming call, or normal device behavior. A fast answer may reflect prior familiarity. Do not auto-fail a student based on one such signal.

Preserve due process in the product

Keep suspected attempts scored but mark comparative eligibility pending when a documented policy requires review. Offer a human review trail and a student explanation route. Private practice should not require intrusive webcam surveillance. If institutional proctoring is added later, scope it as a separate reviewed product with explicit consent, data retention, accessibility exceptions, and operational ownership. Make fullscreen an optional rehearsal convenience unless an approved institution policy specifies otherwise, and keep all warnings calm and factual.

ACCEPTANCE / PRECAUTIONS

- The running response payload contains no answer-key or solution data.
- An integrity flag cannot silently erase an accepted answer or score.
- Comparison exclusion has a documented reason and review path.

Acceptance matrix and adversarial rehearsal

ENGINE RELEASE CHECKS

Prove behavior with realistic journeys and boundary cases before scaling public access. The goal is not a large test count; it is evidence that the engine protects the answer ledger, applies the declared policies, and makes limitations visible to the student.

Cover each examination journey

Run a standard JEE paper with numeric zero and negative values, unsaved palette navigation, both review states, and expiry. Run BITSAT with one missing base answer, successful extra entry, double-click entry, a stale base edit, and refresh inside the locked phase. Run both WBJEE papers, a partial-credit set, an incorrect added option, a paper-only report, and irreversible-bubble mode. Run COMEDK with correct, wrong, and blank responses, verifying the timing evidence blocker.

Keep each scenario's fixtures and expected outcomes in the repository. Test the source-policy configuration independently from UI rendering. Make a compact release report show scenario name, expected behavior, actual result, application version, and run date. A passing demo with placeholder scoring is not acceptable evidence.

Exercise persistence and timing failures

Inject a lost acknowledgement, delayed response, duplicate command, stale question revision, server restart after persisted save, unavailable local storage, and browser sleep past deadline. Race submit against expiry and extra-phase entry against expiry. Test whether a slow worker drains accepted ingress records before snapshot closure. Ensure retries reuse command IDs and final receipts. Observe failures in server logs without recording unnecessary sensitive content.

Check accessibility and release boundaries

Complete one full attempt with keyboard only, screen-reader navigation, 200 percent zoom, and a small mobile viewport. Verify answers survive layout changes, and warnings do not seize focus. Try reading another student's attempt, requesting solutions before release, editing a submitted response, and changing a locked BITSAT base answer. Require zero unexplained score differences, no lost acknowledged answers, and no silent extra time before approving release. Keep performance-load validation in Volume 05, using this same behavioral suite under load.

ACCEPTANCE / PRECAUTIONS

- All four exam paths pass deterministic item and aggregate score checks.
- The attempt is recoverable after a server or browser restart.
- No unresolved release blocker is hidden behind a polished interface.

Sources, evidence boundaries, and builder handoff

PRIMARY-SOURCE REGISTER

Use these source IDs in the policy registry and reviewer report. The documents establish examination facts; the session architecture, persistence rules, and interface safeguards in this volume are proposed product requirements. They are not claims that the examination authorities use the same software implementation.

Record the verified primary documents

[S2-01] NTA, JEE Main Information Bulletin 2026, linked by the current official site and stored in a November 2025 upload path. Reviewed sections: Paper 1 scheme, numerical entries, marking anomalies, duration table, and CBT appendix. [S2-02] BITS Pilani, BITSAT-2026 brochure, retrieved from the official admissions domain. Reviewed format, marking, language, extra-question conditions, and the base-answer lock. The original issuance date is not asserted.

[S2-03] WBJEEB, Information Bulletin WBJEE-2026, official 10 March 2026 upload. Reviewed paper schedule, category distribution, OMR restrictions, and scoring methodology. [S2-04] COMEDK UGET Information Brochure 2026, notified on 3 February 2026, version 1.0. [S2-05] COMEDK current About UGET and Notification 2026 page. Reviewed counts, marks, language, CBT controls, and FAQ response revision guidance. All five were checked on 11 October 2026.

Carry unresolved evidence forward

COMEDK's reviewed current primary documents do not explicitly establish a numeric 2026 duration. Keep the proposed 180-minute custom-mock setting unresolved until reviewed official timing evidence is attached. Source navigation may lead to older schedules; inspect the document year, not merely the current page heading. Exact BITSAT and COMEDK save-control microbehavior should be reviewed against current official interface evidence before using an exact-parity claim.

Return a reviewable handoff

Deliver the implemented policy JSON, field-level source mapping, evidence gaps, deterministic scoring fixtures, transition diagram, API contracts, and acceptance report. Link sources from the administrator policy view. Preserve a source-document checksum where lawfully obtained so the reviewed edition can be identified. Recheck rules before a new examination-year release and create a new policy version for changes. Carry custom training decisions separately; a useful independent adaptation does not need a false claim of official endorsement.

ACCEPTANCE / PRECAUTIONS

- Source IDs map to working primary URLs in the appended register.
- The selected examination year is visible before starting.
- Unresolved evidence remains a release blocker for the corresponding verified claim.

Source register and implementation handoff

EVIDENCE AND SCOPE OF THIS SPECIFICATION

Use these references to verify the stated baseline. Requirements and engineering thresholds in this volume are proposed product decisions unless a source explicitly establishes them.

S2-01 | NTA JEE Main Information Bulletin 2026

https://cdnbbsr.s3waas.gov.in/s3f8e59f4b2fe7c5705bf878bbd494ccdf/uploads/2025/11/20251102 1649722475.pdf

Checked: 2026-10-11. Official current-site linked 2026 bulletin. Source upload path is November 2025; original publication date not independently asserted. Printed pages 11-16 and 59-64 support relevant format, scoring, duration, and CBT rules.

S2-02 | BITS Pilani BITSAT-2026 brochure

https://www.bitsadmission.com/FD/downloads/BITSAT-2026_brochure.pdf

Checked: 2026-10-11. Official admissions brochure, document year 2026. Original issuance date not asserted. PDF pages 6-8 describe format, extra questions, language, and score reporting.

S2-03 | WBJEEB Information Bulletin WBJEE-2026

https://cdnbbsr.s3waas.gov.in/s3d2a27e83d429f0dcae6b937cf440aeb1/uploads/2026/03/2026031 01506582412.pdf

Checked: 2026-10-11. Official upload path dated 10 March 2026. Printed pages 8-10 cover paper schedule, counts, categories, OMR restrictions, and scoring.

S2-04 | COMEDK UGET 2026 Information Brochure, version 1.0

https://www.comedk.org/uploads/Information-brochure-2026-version-1.0.pdf

Checked: 2026-10-11. Notified 3 February 2026. Printed pages 22-23 and 29 establish counts, marking, language, CBT facilities, and response revision. Numeric 2026 duration not explicitly found; simulator release requires separate timing evidence.

S2-05 | COMEDK About UGET and Notification 2026

https://www.comedk.org/about-uget-and-notification-2026

Checked: 2026-10-11. Current official exam page corroborates 2026 structure, marking, language, and response revision. Does not explicitly state numeric duration in retrieved text. Do not use older important-date destinations as 2026 duration evidence.

ACCEPTANCE / PRECAUTIONS

- Before release, retain the approved configuration, source document date, verification owner and test evidence. Re-check any rule that changes for a new exam year.